Associate Automation Engineer curriculum for the “Veeva integrated curriculum program for college grads”

Focus Area	Topics	Hands-on Skills
1. Testing Foundation - SDLC - Fundamentals of Testing	**SDLC** ❖ Requirements Design ❖ Coding ❖ Testing ❖ Deployment **Fundamentals of Testing** ❖ Quality Assurance ❖ System Integration Testing ❖ User Acceptance Testing ❖ Agile vs Waterfall ❖ Bug Tracking Tool (JIRA) usage ❖ Test Case Management (XRay) usage ❖ Confluence usage	➢ Set up a development environment using Intellij IDE. ➢ Basic Java programming hands-on leveraging the concepts mentioned ➢ Understand test life cycle using XRay - create and execute tests
2. Core Java - Basics - OOPS - Collections - Streams - Lambda Expressions - Exception Handling - Java I/O Operations	**Basics** ❖ Class, Object and package ❖ Data Types ❖ Control Flow ❖ Methods ❖ Code debugging ❖ Loops, Arrays **OOPS** ❖ Encapsulation ❖ Inheritance ❖ Polymorphism ❖ Abstraction **Collections** ❖ Lists ❖ Set ❖ Map **Streams** ❖ Intermediate operations	➢ Apply OOPS principles to build simple, modular Java classes. ➢ Basic Java programming hands-on leveraging the concepts mentioned ➢ Use appropriate collections for different scenarios. ➢ Perform data manipulation with Streams. ➢ Implement robust error handling ➢ Write concise, functional code using Collections and Lambdas. ➢ Write code to perform CRUD operations on files.

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	❖ Terminal operations **Lambda Expressions** ❖ Syntax & use cases **Exception Handling** ❖ try, catch, finally ❖ Throw/ Throws ❖ Custom exceptions **Java I/O Operations** ❖ Read and Write File ❖ Creation of File ❖ Buffer Reader ❖ Buffer Writer ❖ File Permissions	
3. Advanced Java - Design Patterns	**Design Patterns** ❖ Singleton ❖ Builder Pattern	➢ Build a simple test automation configuration manager. ➢ Use Singleton for ConfigurationManager ➢ Use Builder for TestConfig ➢ Read configuration values such as browser, URL, timeout etc. ➢ Create multiple TestConfig objects using Builder. ➢ Ensure only one ConfigurationManager instance exists
4. Build Tools - Maven	**Maven** ❖ Project structure ❖ Maven lifecycle ❖ Goals ❖ Scope of each Goal ❖ Dependency Management ❖ Plugins configuration ❖ Settings.xml file ❖ Profiles ❖ Nexus Repository config	➢ Create multi-module Maven projects managing project dependencies ➢ Create different maven profiles ➢ Configure surefire, compiler, fail safe, etc. plugins in pom.xml file ➢ Setup local nexus repository ➢ Configure Settings.xml file to access the artifacts from local nexus repository
5. Version Control - Git Operations	**Git Operations** ❖ Repo cloning via ssh ❖ Repo cloning via https ❖ Difference ssh vs https clone ❖ Clone Repo ❖ Fetch and Pull	➢ Create a local git repository and push it to remote ➢ Create a branch using git command line and push it remote ➢ Rebase a local branch with remote

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	❖ Commit and Push ❖ Branching ❖ Merge and Rebase ❖ Commit diff	➢ Commit message using git command line ➢ Pull latest code from remote branch and fix merge conflicts
6. Test Automation - Selenium - Playwright - REST API Basics - Postman - API Automation	**Selenium** ❖ Webdriver setup. Browser initialization ❖ Locating elements (DOM structure) ❖ CSS Selectors ❖ Webdriver Commands ➢ Navigation ➢ Element interaction ❖ Synchronization ➢ Implicit vs Explicit vs Fluent waits ❖ Verification & Validation Actions ❖ Mouse & Keyboard Actions ❖ Dropdown Actions ❖ Screenshot Capture ❖ Windows and Frame Handling ❖ Multi Tab handling ❖ Page Object Model (POM) **Playwright** ❖ What is a Playwright? ❖ Playwright vs Selenium ❖ Install Playwright ❖ Playwright supported tech stack ❖ Cross Brower testing ❖ Handing Web Elements. ❖ Waits & Synchronization **REST API Basics** ❖ REST Architecture ❖ REST API Principles ❖ Status codes **Postman** ❖ Request building ➢ GET ➢ POST ➢ PUT ➢ DELETE	➢ Browser initialization and locate DOM elements ➢ Write robust web element locators. ➢ Perform basic user actions on a web page ➢ Implement effective waits to handle dynamic pages ➢ Develop reliable test steps ➢ Create a complete automation framework (Page Object Model) using all learned tools. ➢ Setup node.js and Playwright in local machine ➢ Create a playwright project ➢ Understand project folder structure ➢ Understand playwright.config ➢ Run first UI playwright test ➢ Run the tests from command line ➢ Manually test API using Postman ➢ Write basic Postman test assertions.

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	❖ Environment variables ❖ Basic scripting (pre-request/ Tests) **API Automation** ❖ REST Assured ➢ Setup ➢ Building requests (params, headers, body) ❖ Authentication ➢ Basic/ Bearer token ❖ Response validation ➢ JSON/ XML parsing ➢ Schema validation	➢ Automate CRUD operations with REST Assured ➢ Integrate API tests into the Maven build. ➢ Create a post api call with basic authentication ➢ Read the Post call body from the Json file. ➢ Perform GET API schema validation against the response
7. Testing Frameworks - TestNG	**TestNG** ❖ Annotations ❖ Grouping ❖ Parameterization ❖ Data Providers ❖ Listeners ❖ Assertions ➢ Soft ➢ Hard	➢ Convert existing Selenium tests to TestNG ➢ Read the data from csv file in Data provider ➢ Create a group of regression Test cases and run it ➢ Debug test case failures
8. BDD - Gherkin - Cucumber	**Gherkin** ❖ Given ❖ When ❖ Then ❖ And ❖ * **Cucumber** ❖ Feature files ❖ Scenario ❖ Scenario Outline ❖ Background ❖ Rule ❖ Step definitions ❖ Runner File ❖ Tags ❖ Cucumber Options ❖ Hooks	➢ Implement BDD feature files using Cucumber. ➢ Write reusable step definitions ➢ Write different test cases using Scenario and Scenario outline ➢ Create cucumber hooks class ➢ Configure cucumber options in runner class

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9. DevOps - DevOps Basics - Jenkins	**DevOps Basics** ❖ Phases of DevOps lifecycle **Jenkins** ❖ Setup Jenkins in local machine ❖ Setting up CI/CD pipelines ❖ Cron Schedules ❖ Pipeline stages ❖ Parallel stages ❖ Stage dependency ❖ Trigger types ❖ SCM polling ❖ Webhooks ❖ Debugging pipeline ❖ Failures & Common issues ❖ Log analysis	➢ Install Jenkins locally ➢ Configure Java/JDK ➢ Access Jenkins dashboard ➢ Create and manage users ➢ Configure basic Jenkins settings ➢ Create a Freestyle project ➢ Configure source code repository ➢ Configure build steps ➢ Configure post-build actions ➢ Run jobs manually ➢ Analyze and fix a failed test case.
10. Cloud Fundamentals - AWS	**AWS** ❖ Accessing AWS ❖ Management console ❖ Command line interface (CLI) ❖ S3, EC2, EBS, AMI, Lambda ❖ Other elastic computes (ECS, EKS, ECR)	➢ Register AWS account ➢ Create EC2 machines for Linux and Windows. ➢ Create S3 buckets ➢ Store data in S3 bucket ➢ Configure object permissions ➢ Enable bucket versioning ➢ Create an EBS volume ➢ Attach an EBS volume to EC2 ➢ Detach and reattach an EBS volume ➢ Understand EBS availability-zone dependency
11. AI Fundamentals - Gemini - Claude	**Gemini Web** ❖ Basics ❖ Writing good Prompts ❖ Creating GEMS **Claude** ❖ Basics ❖ Writing good Prompts ❖ Using Claude CLI ❖ Creating md files, skills ❖ Creating agents and MCPs ❖ Integration with Playwright	➢ Create a Gemini GEM for repetitive tasks. ➢ Create a Claude skill for repetitive tasks using CLI ➢ Create, read, and edit md files with Claude ➢ Generate and refactor automation code using Claude ➢ Generate framework documentation using md file ➢ Connect Claude to an MCP server ➢ Build a custom MCP server using Claude API

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		➢ Create a new Claude agent for your automation project ➢ Design end to end Claude workflow using skills, agents, MCP tools for your automation project ➢ Create UI automation scripts using Playwright MCP
12. Database - RDBMS	**RDBMS** ❖ Relational DB concepts ❖ SQL (Structured query language) ➢ DDL ➢ DML ➢ DCL ➢ TCL ❖ Database design & structure ➢ Data Types ➢ Constraints ➢ Normalization ❖ Querying and data retrieval ❖ Data Manipulation	➢ Query database to fetch results from a Table ➢ Query database to fetch results from multiple tables using joins (Optimized search) ➢ Update the table using key columns
13. Linux - Linux basics	**Linux** ❖ Basic Linux Commands ❖ Linux File System ❖ File Management in Linux ❖ File Permissions	➢ Find a file in Linux using a command ➢ Change file permission ➢ Check CPU memory and disk usage ➢ Read logs and understand the root cause ➢ Create and extract .tar.gz files ➢ grep a term (word) and list the files that has the word occurrences ➢ Create and execute basic shell script